SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0304	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	SAMUEL K	**Reg. No.:**	192424063
Section / Batch:	B. TECH AIDS	**Date:**	20/10/2026

Code of Conduct

*I _____**SAMUEL K**_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____SAMUEL K_____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

- Develop a Python program to implement a rule-based morphological processing system for the input words "analyzing", "analysis", and "analytical". The program should identify the root word and separate the corresponding prefixes or suffixes using suitable morphological rules, classify each word as an inflectional or derivational form, normalize all related variants to a common representation for indexing, and generate a structured report showing the original word, extracted root, identified affix(es), transformation type, and normalized output to support efficient search engine retrieval.

CODE:

```
import re

# Embedded dataset
words = ["analyzing", "analysis", "analytical"]

def morphological_analysis(word):
    word = word.lower()

    # Rule 1: Inflectional form of analyze
    if re.search(r"ing$", word):
        root = re.sub(r"ing$", "", word) + "e"
        affix = "-ing"
        form_type = "Inflectional"
        normalized = "analyze"

    # Rule 2: Irregular related lexical form
    elif word == "analysis":
        root = "analysis"
        affix = "Lexical ending: -sis"
        form_type = "Related lexical noun"
        normalized = "analyze-family"

    # Rule 3: Derivational adjective
    elif re.search(r"ical$", word):
        root = re.sub(r"ical$", "ic", word)
        affix = "-al"
        form_type = "Derivational"
        normalized = "analyze-family"

    else:
        root = word
        affix = "None"
        form_type = "Base form"
        normalized = word

    # Common family-level representation
```

```
    if word in ["analyzing", "analysis", "analytical"]:
        normalized = "ANALYZ-family"

    return root, affix, form_type, normalized


print(f"{'Word':<15}{'Root/Base':<20}{'Affix':<25}"
      f"{'Type':<25}{'Normalized'}")

print("-" * 105)

for word in words:
    root, affix, form_type, normalized = morphological_analysis(word)

    print(f"{word:<15}{root:<20}{affix:<25}"
          f"{form_type:<25}{normalized}")
```

OUTPUT:

Word	Root/Base	Affix	Type	Normalized
analyzing	analyze	-ing	Inflectional	ANALYZ-family
analysis	analysis	Lexical ending: -sis	Related lexical noun	ANALYZ-family
analytical	analytic	-al	Derivational	ANALYZ-family

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

- Develop a Python program to create a morphological parser for the input words "disagree", "agreement", and "agreeable". The program should detect and separate prefixes, suffixes, and the base word using rule-based morphological analysis, examine how derivational modifications alter the meaning of the root word, classify each transformation according to its morphological type, and produce a comprehensive output containing the original word, identified prefix, root word, suffix, transformation category, semantic interpretation, and normalized base form for use in sentiment analysis applications.

CODE:

```
import re

# Embedded dataset
words = ["disagree", "agreement", "agreeable"]

for word in words:
    prefix = ""
    suffix = ""
    root = word
    meaning = ""
    category = ""

    if re.match(r"^dis", word):
        prefix = "dis-"
```

```
        root = re.sub(r"^dis", "", word)
        category = "Prefix derivation"
        meaning = "Negation or opposition"

    elif re.search(r"ment$", word):
        suffix = "-ment"
        root = re.sub(r"ment$", "", word)
        category = "Noun-forming derivation"
        meaning = "State or result of agreeing"

    elif re.search(r"able$", word):
        suffix = "-able"
        root = re.sub(r"able$", "", word)
        category = "Adjective-forming derivation"
        meaning = "Pleasant or acceptable"

    print(f"\nWord: {word}")
    print(f"Prefix: {prefix if prefix else 'None'}")
    print(f"Root: {root}")
    print(f"Suffix: {suffix if suffix else 'None'}")
    print(f"Transformation: {category}")
    print(f"Meaning: {meaning}")
    print("Normalized base: agree")
```

OUTPUT:

```
Word: disagree
Prefix: dis-
Root: agree
Suffix: None
Transformation: Prefix derivation
Meaning: Negation or opposition
Normalized base: agree

Word: agreement
Prefix: None
Root: agree
Suffix: -ment
Transformation: Noun-forming derivation
Meaning: State or result of agreeing
Normalized base: agree

Word: agreeable
Prefix: None
Root: agree
Suffix: -able
Transformation: Adjective-forming derivatior
Meaning: Pleasant or acceptable
Normalized base: agree
```

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

- Develop a Python program to implement a morphology-based normalization module for the input words "govern", "government", and "governance". The program should perform morphological decomposition to identify the root word and successive derivational suffixes, determine the derivational level associated with each word, normalize all variants to a common lexical representation, and generate a structured output displaying the original word, root form, detected affix(es), derivational hierarchy, normalized representation, and the final output used for topic modeling and document clustering.

CODE:

```
import re

# Embedded dataset
words = ["govern", "government", "governance"]

for word in words:
    if re.search(r"ment$", word):
        root = re.sub(r"ment$", "", word)
        suffix = "-ment"
        level = 1

    elif re.search(r"ance$", word):
        root = re.sub(r"ance$", "", word)
        suffix = "-ance"
        level = 1

    else:
        root = word
        suffix = "None"
        level = 0

    print(f"\nWord: {word}")
    print(f"Root: {root}")
    print(f"Suffix: {suffix}")
    print(f"Derivational level: {level}")
    print("Normalized representation: GOVERN-family")
```

OUTPUT:

```
Word: govern
Root: govern
Suffix: None
Derivational level: 0
Normalized representation: GOVERN-family

Word: government
Root: govern
Suffix: -ment
Derivational level: 1
Normalized representation: GOVERN-family

Word: governance
Root: govern
Suffix: -ance
Derivational level: 1
Normalized representation: GOVERN-family
```

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation.
Derive the morphological structure and normalized root for each input word.

- Develop a Python program to implement a morphological parsing and normalization system for the input words "activate", "activation", and "reactivation". The program should detect and separate prefixes, root words, and suffixes using rule-based morphological parsing, identify the sequence of derivational transformations that produce each word, analyze the effect of each morphological modification on the word class and semantic meaning, and generate a structured report showing the original word, prefix (if present), root word, suffix, derivational sequence, normalized base form, and final parsed representation for use in document classification and semantic indexing.

CODE:

```
import re

# Embedded dataset
words = ["activate", "activation", "reactivation"]

for word in words:
    prefix = ""
    suffix = ""
    root = word

    if re.match(r"^re", word):
        prefix = "re-"
        root = re.sub(r"^re", "", word)

    if re.search(r"ion$", root):
```

```
        suffix = "-ion"
        root = re.sub(r"ion$", "", root)

        # Restore the base verb for activation/reactivation
        if root == "activat":
            root = "activate"

    elif word == "activate":
        root = "activ"
        suffix = "-ate"

    if word == "activate":
        word_class = "Verb"
        meaning = "To make active"
    elif word == "activation":
        word_class = "Noun"
        meaning = "Process or state of activating"
    else:
        word_class = "Noun"
        meaning = "Process of activating again"

    print(f"\nWord: {word}")
    print(f"Prefix: {prefix if prefix else 'None'}")
    print(f"Root/base: {root}")
    print(f"Suffix: {suffix if suffix else 'None'}")
    print(f"Word class: {word_class}")
    print(f"Meaning: {meaning}")
    print("Normalized representation: ACTIVATE-family")
```

OUTPUT:

```
Word: activate
Prefix: None
Root/base: activ
Suffix: -ate
Word class: Verb
Meaning: To make active
Normalized representation: ACTIVATE-family

Word: activation
Prefix: None
Root/base: activate
Suffix: -ion
Word class: Noun
Meaning: Process or state of activating
Normalized representation: ACTIVATE-family

Word: reactivation
Prefix: re-
Root/base: activate
Suffix: -ion
Word class: Noun
Meaning: Process of activating again
Normalized representation: ACTIVATE-family
```

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word.

- Develop a Python program to implement an inflectional morphology-based normalization module for the input words "create", "creates", and "creating". The program should recognize grammatical suffixes corresponding to different inflectional forms, identify the underlying root word by applying appropriate normalization rules, classify each input according to its grammatical feature (e.g., base form, third-person singular, present participle), and produce a structured output containing the original word, identified suffix, grammatical category, extracted root, normalized base form, and the final normalized representation to support efficient search optimization and information retrieval.

CODE:

```
import re

# Embedded dataset
words = ["create", "creates", "creating"]

for word in words:
    if re.search(r"s$", word) and word != "creates":
        root = re.sub(r"s$", "", word)
        suffix = "-s"
        category = "Third-person singular present"

    elif re.search(r"s$", word):
        root = re.sub(r"s$", "", word)
        suffix = "-s"
        category = "Third-person singular present"
```

```
    elif re.search(r"ing$", word):
        root = re.sub(r"ing$", "", word) + "e"
        suffix = "-ing"
        category = "Present participle / gerund"

    else:
        root = word
        suffix = "None"
        category = "Base form"

    print(f"\nWord: {word}")
    print(f"Suffix: {suffix}")
    print(f"Grammatical category: {category}")
    print(f"Root/lemma: {root}")
    print(f"Normalized form: create")
```

OUTPUT:

```
Word: create
Suffix: None
Grammatical category: Base form
Root/lemma: create
Normalized form: create

Word: creates
Suffix: -s
Grammatical category: Third-person singular presen
Root/lemma: create
Normalized form: create

Word: creating
Suffix: -ing
Grammatical category: Present participle / gerund
Root/lemma: create
Normalized form: create
```

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues

		relevant issues and factors involved			
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____